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Octane IDE Bug & False Graphics Failures

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Revision History

REV	DATE	REASON FOR CHANGE
Rev 0	01/29/99	Initial Release

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Subject:

Silicon Graphics Octane Hardware Diagnostics (IDE) And New ESI Graphics Boards.

Problem:

SGI Integrated Development Environment (IDE diagnostics) may produce false graphics test failures when run on newer SGI Octanes with ESI graphics boards (started shipping FW3/99). The SGI IDE diagnostics version is printed at the top of the screen when they are booted/started. Version dated Feb 5, 1997 has a known bug that results in the intermittent false ESI graphics error report.

Solution:

Ignore IDE graphics test failures (only) when running Octane console diagnostics on systems containing the SGI ESI graphics boards. The QX/i M4 software release and CT/i 6.0 software FMI in 1999 will include corrected versions of IDE diagnostics that will no longer have this problem. If system dual CRT screens are operating normally then the graphics boards are fine.

Supporting Information:

To identify whether or not your particular system OC Octane has the new SGI ESI graphics boards or not, use the Irix 'hinv' command ('hinv' shows SGI hardware inventory found at OC bootup). The newer SGI ESI graphics boards are identified as "ESI" as shown in bold underline in the below 'hinv' output example. If your system OC Octane has these boards, the SGI Feb 5, 1997 IDE hardware diagnostics may report false errors during the graphics (gfx) portion of the diagnostics. An example of a typical failure report is included after the "hinv" example.

Note: To run "hinv", open a shell from the service desktop and type **hinv** followed by pressing the return button. The next page has an example of the output.

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Example of output for "hinv" for both ESI and SI systems.

Octane computer with new ESI graphics boards	Octane computer with SI graphics boards
1 225 MHZ IP30 Processor CPU: MIPS R10000 Processor Chip Revision: 3.4 FPU: MIPS R10010 Floating Point Chip Revision: 0.0 Main memory size: 256 Mbytes Instruction cache size: 32 Kbytes Data cache size: 32 Kbytes Secondary unified instruction/data cache size:1Mbyte Integral SCSI controller 0: Version QL1040B (rev. 2), single ended Disk drive: unit 1 on SCSI controller 0 Disk drive: unit 2 on SCSI controller 0 Integral SCSI controller 1: Version QL1040B (rev. 2), single ended Optical disk: unit 3 on SCSI controller 1 Comm device: unit 4 on SCSI controller 1 Comm device: unit 4, lun 1 on SCSI controller 1 CDROM: unit 6 on SCSI controller 1 IOC3 serial port: tty1 IOC3 serial port: tty2 IOC3 parallel port: plp1 <u>Graphics board: ESI with texture option</u> <u>Graphics board: ESI</u> Integral Fast Ethernet: ef0, version 1, pci 2 Fast Ethernet: ef1, version 1, pci 3 Iris Audio Processor: version RAD revision 12.0, number 1	1 195 MHZ IP30 Processor CPU: MIPS R10000 Processor Chip Revision: 2.7 FPU: MIPS R10010 Floating Point Chip Revision: 0.0 Main memory size: 256 Mbytes Instruction cache size: 32 Kbytes Data cache size: 32 Kbytes Secondary unified instruction/data cache size:1Mbyte Integral SCSI controller 0: Version QL1040B (rev. 2), single ended Disk drive: unit 1 on SCSI controller 0 Disk drive: unit 2 on SCSI controller 0 Integral SCSI controller 1: Version QL1040B (rev. 2), single ended Optical disk: unit 3 on SCSI controller 1 Comm device: unit 4 on SCSI controller 1 Comm device: unit 4, lun 1 on SCSI controller 1 CDROM: unit 6 on SCSI controller 1 IOC3 serial port: tty1 IOC3 serial port: tty2 IOC3 parallel port: plp1 Graphics board: SI with texture option Graphics board: SI Integral Fast Ethernet: ef0, version 1, pci 2 Fast Ethernet: ef1, version 1, pci 3 Iris Audio Processor: version RAD revision 12.0, number 1

Example of Octane IDE bug and false graphics error message*This may happen at during the boot up or running IDE's.*

```

bkend FAILED
failure detected on gfx in XI0 slot 12
part 030-1241-002 type MOT10
serial HRL585 rev G

```

Other IDE tests that may show false failures:

It is possible, according to a SGI bug report, that other graphics tests besides the 'bkend' test may also intermittently fail. These other gfx tests include:

```

rdram_mem_quick
dmatest
mg_test_tra

```

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Example of diagnostic test panic or crash:

In some situations, a diagnostic test panic/crash has also been known to occur. A typical IDE panic/crash is shown below for reference. It may take other forms, the key words below are "PANIC: Unexpected exception"...

```
Exception: < vector=Normal>
Status register: 0x34050082 <CU1, CU0, FR, CH, DE, IPL=8, KX, MODE=KERNEL>
Cause register: 0x801c(E=0, IP8, EXC=DBE)
Exception PC: 0x8000000208a24d4, Exception RA: 0x8000000208a24c8
Data Bus Error
XBOW status: 0x1000000 <LINK8>
LINK 0x8 status: 0xffffffff80060000
<LINK_ALIVE, MULTI, ILLEGAL_DST, BANDWIDTH_ALLOC_PORT=0x0>
Aux status: 0x20<RCV_RETRIES=0,
XMT_RETRIES=0, TIMEOUT_SRC=0x0, PORT=ON, PORT_SIZE=16>
Error command word: 0x0
Error address: 0x0
IOC3 PCI status register: 0xa800146<SIG_TAR_ABRT>
VID 0's saved user regs in hex (gpda=0x800000020a28230
arg: 1 10000000 a80000002abf428 1
tmp: (not captured)
svo: (not captured)
t8: (not captured)
v0: (not captured)
gp: (not captured)
PANIC: Unexpected exception
```

What should be done if these false failures are noticed?

Any IDE PANIC of this form, on any version or configuration, is likely NOT a hardware failure but an IDE diagnostic software bug. The best thing to do to make sure this is not a problem is to continue to bring the system up and operate the scanner in normal operation. If the monitors are operating normally and the graphics seem normal, there is no evidence to replace the graphics boards.

When is this situation likely to be fixed?

The IDE version and false problems described are expected to be corrected as of the LightSpeed QX/i M4 release and in the CT/i 6.0 FMI later this year. A new and corrected version of SGI IDE will be included in those releases.



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